We now have an `employees` index with the expected schema mapping. In this example, we dictated the types to Elasticsearch; now we are in control of the schema.

Did you notice the `address` field in listing 4.4? It is an `object` type consisting of additional fields: `street` and `country`. However, the `address` field type has not been mentioned as `object`, although we said it was an `object` that encapsulates other data fields. This is because the `object` data type is deduced by Elasticsearch for any inner objects by default—hence we didn't mention it explicitly. Also, the subfield `properties` object wrapped in the `address` helps define further attributes of the inner object.

Now that our `employees` index is in production, suppose we are given a new requirement: management wants us to extend the model to include a few more attributes such as department, phone number, and others. To fulfill this requirement, we must add these fields using the mapping API on a live index. We go over this topic briefly in the next section.

4.3.2 Updating schema using the mapping API

As our project matures, the data model will undoubtedly change, too. Continuing with our employee document, let's add `joining_date` and `phone_number` attributes:

```
{
  "name":"John Smith",
  "joining_date":"01-05-2021",
  "phone_number":"01234567899"
  ...
}
```

The joining date is a `date` type as we want to do date-related operations, such as sorting employees by this date. The phone number is expected to be stored as is (we don't want the data to be tokenized—more on this in the coming sections), so it fits the `keyword` data type. To amend the existing `employees` schema definition with these additional fields, we invoke a `_mapping` endpoint on the existing index, declaring the new fields in the request object.

Listing 4.5 Updating the existing index with additional fields

```
PUT employees/_mapping
{
  "properties":{
    "joining_date":{
      "type":"date",
      "format":"dd-MM-yyyy"
    },
    "phone_number":{
      "type":"keyword"
    }
  }
}
```

_mapping endpoint to update the existing index

The joining date field is a date type.

Expected date format

Phone numbers are stored as is.